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Subject - STLD (Switching Theory And Logic Design).

Subject code - ETEC-205

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Answers

Ans 1: (a) - 100 1011 to gray code.

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• Advantage of using gray code:

- Advantage of gray code over binary is that only one bit changes for each step. Because of this feature, an amount of switching is minimised and the reliability of switching system is improved.
- low power consumption & it also results in fast response.
- this code is used for shaft encoder.

①

Teacher's Sign.....

(b)

combinational circuit

- ① output of a combinational circuit at any instant of time only depends upon the ~~input~~ present input.

- ② There is no memory unit

- ③ There is no clock

- ④ faster in speed

~~c) $(xy)' + z' + z + xy + xw$~~

eg. Half adder, full adder, multiplexer, de-multiplexer, encoder and decoder.

sequential circuit.

- ① output of a sequential circuit at any given instant ~~time is known as~~ ~~the present state~~ depends upon present input and past output - and sequence in which input is applied

- ② There is a memory unit to store immediate ~~past~~ output.

- ③ there is clock

- ④ slower than combinational circuit

eg. Flip flop and registers

$$c) [(xy)' + z']' + z + xy + xw$$

$$= (xy) \cdot z' + z + xy + xw \quad (\text{using demorgan's law})$$

$$= xy(z' + 1) + xw + z$$

$$= \boxed{xy + z + xw} \quad | \text{ as } A + \bar{A}B = A + B$$

②

Teacher's Sign.....

d1. Ring counter.

It is a type of ring counter in which output of last flip-flop is connected to the input of first flip-flop. It circulates a single '0' or '1' bit around the ring.

Twisted ring counter.

Twisted ring counter is a type of ring counter in which complement of output of last shift register is connected to input of the first register. It circulates a stream of 1s followed by 0s around the ring.

Ans. (c). ① Fan out.

This is number of similar gates which can be driven by a gate. High fan out is advantageous because it reduces the need for additional drivers to drive more gates.

② Shift Register

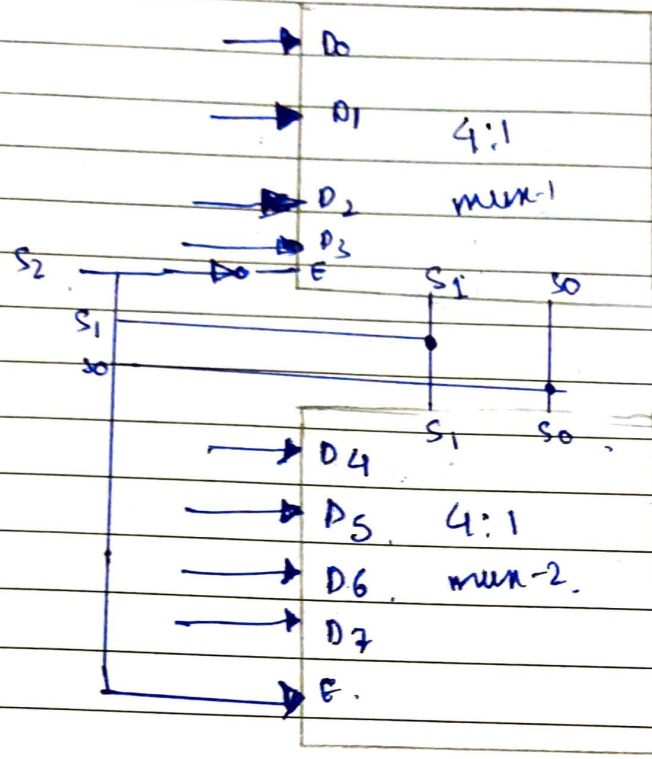
A register capable of shifting its binary information either to right or to the left is called a shift register. These allow data transfer from one flip flop to the other or outside it with the application of clock pulse.

Shift register find application for data storage, data transfer, & certain arithmetic and logic operations.

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Ans. a(a).

8:1 mux using 4:1 mux



truth table.

S_2	S_1	S_0	Output
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

2. (b.)

AB \ CD	00	01	11	10
00	1	1	1	1
01			1	1
11	1		x	1
10		x	x	x

$$f(A, B, C, D) = A' (B' C + B' C + B C D) + A D' (C + A)$$

$$d(A, B, C, D) = A B' (C' D + C D') + A C D$$

$$f(A, B, C, D) = A B' C' + A' B' C + A' B C D + B D' C + B D' A$$

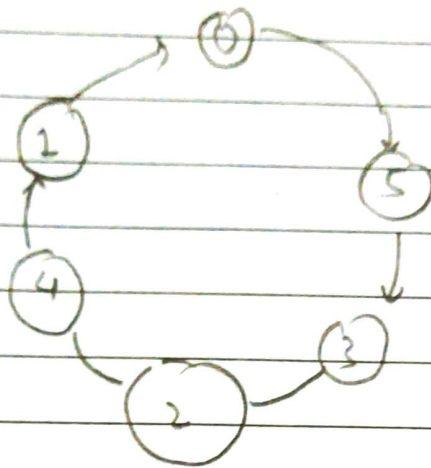
$$= A' B' C' (D + D') + A' B' C' (D + D') + A' B C D + (A + A') B D' C + A B D' (C + C')$$

$$= A' B' C' D + A' B' C' (D + D') + A' B C D + (A + A') B D' C + A B D' (C + C')$$

$$d(A, B, C, D) = A B' C' D + A B' C' D + A C D B + A C D B'$$

Ans 3(a) mod 6 synchronous counter.

This counter has only six stable states, but it requires flip flops.



mod number

$$= 2^n$$

$$= 2^3$$

<u>PS</u>			<u>NS</u>			<u>Required excitation</u>		
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	T_3	T_2	T_1
0	0	0	1	0	1	1	0	1
1	0	1	0	1	1	1	1	0
0	1	1	0	1	0	0	0	1
0	1	0	1	0	0	1	1	0
1	0	0	0	0	1	1	0	1
0	0	1	0	0	0	0	0	1

$Q_3 \backslash Q_2 Q_1$	00	01	11	10
0	1	-	-	1
1	1	1	1	x

$$T_3 = \bar{Q}_1 + Q_3$$

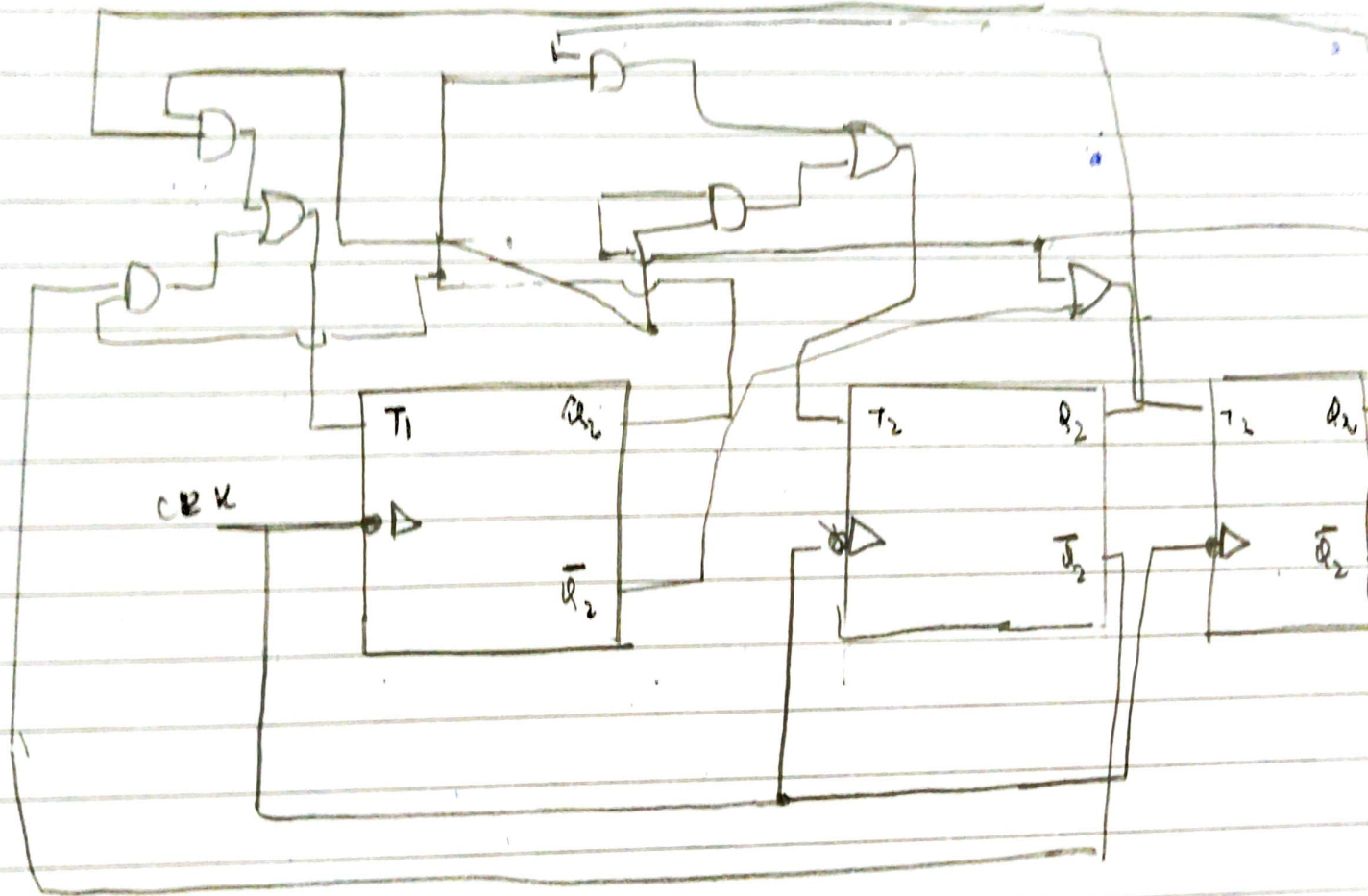
$Q_3 \backslash Q_2 Q_1$	00	01	11	10
0	-	-	-	1
1	-	1	x	x

$$T_2 = Q_1 Q_2 + \bar{Q}_1 Q_2$$

$Q_1 Q_2$

	00	01	11	10
0	1	1	1	
1			x	x

$$T_1 = \overline{Q_1}Q_2 + Q_1\overline{Q_2}$$



Ans (co). JK flip flop is basically a gated SR flip flop with addition of clock input & circuitry that prevents illegal invalid output condⁿ that can occur when both inputs SR are equal to logic level '1'. Due to this additional clocked input, a JK flip flop has 4 possible input combinations - logic '1', logic '0', no change & toggle. The symbol for a JK flip flop is similar to that of an SR bistable latch as seen in previous tutorial.

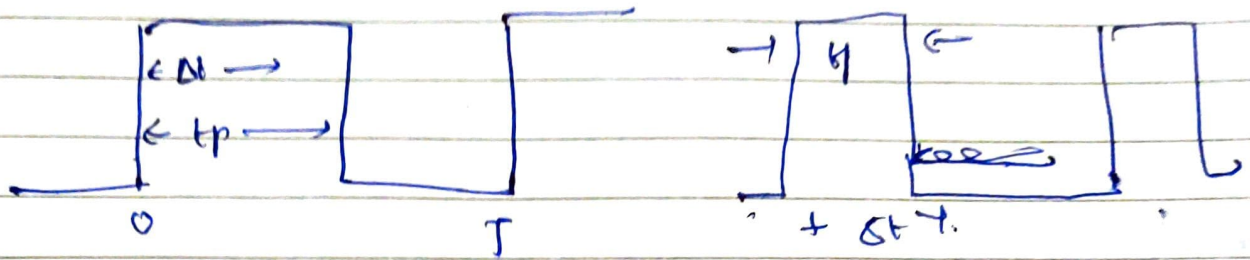
Race around condⁿ - It is important to note that in JK flip flop, the output is feedback to input & therefore change in output results in a change in input, if $J = K = 1$ & $Q = 0$.

When a clock pulse with width τ_p is applied output will change from 0 to 1 after Δt

Δt = delay of 2 level NAND gate
 τ_p = pulse width.

now after Δt , we have $J = K = 1$ & $Q = 1$.
 after another Δt , output will become 0.
 Hence output will oscillate back & forth.

between 0 & 1 in duration of clock pulse width. So at end of clock pulse, value of Q is ambiguous. This is known as race around condⁿ.



It can be removed using

- ① If t_p is ~~less~~ reduced.
- ② If Δt is $\uparrow$ (increased).